

Justin Westley

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EXPERIENCE

Meta

New York, NY

Software Engineer – Wearables – Reality Labs

Feb 2025 – Present

- Engineered embedded firmware and libraries for real-time EMG sensor signal processing and gesture recognition on neural wristbands, used with Meta wearable glasses and future devices.
- Built end-to-end digital signal processing validation frameworks, improving test coverage and model reliability.
- Optimized on-device ML execution with research teams, improving model accuracy and measurement fidelity.
- Designed memory subsystem reducing device footprint by 25% (1000 kB+), enabling larger models.
- Led investigation with ML and research teams resolving numerical discrepancies between theory and on-device models.
- Brought up FVP simulator for next-gen ARM CPU-NPU platform, integrating with the device RTOS and firmware stack, and built CLI tooling enabling ML teams to benchmark models without physical hardware.
- Built framework for running embedded firmware and DSP pipelines on x86 in unit/integration tests, saving ~15 minutes per iteration by eliminating on-device flash and build cycles.

Tech stack: C, C++, Python, TypeScript, Zephyr RTOS, ARM Cortex-M, DSP, On-Device ML, BLE, Buck2, CMake

Amazon Web Services (AWS)

Seattle, WA

Software Development Engineer II – Elastic Compute Cloud (EC2)

Jul 2021 – Oct 2023

- Expanded AWS Edge infrastructure to dozens of global locations, enabling ultra-low-latency compute.
- Owned and operated a highly available service processing 60K+ TPS across AWS's global edge network.
- Built guardrails to prevent customer billing errors and resource cleanup failures.
- Designed distributed auditing system for live networking data, preventing misconfigurations and reducing incidents.
- Migrated legacy PostgreSQL relational store to DynamoDB, improving throughput and resilience.

Tech stack: Java, Ruby, TypeScript, React, Python, Bash, Kotlin, AWS (EC2, DynamoDB, Lambda, SQS, SNS, S3, VPC), PostgreSQL, CloudFormation/CDK

Amazon

Seattle, WA

Software Development Engineer I – Alexa Shopping

Jul 2020 – Jul 2021

- Launched international Prime sign-ups and free-trial flows on Alexa, expanding voice commerce globally.
- Implemented multi-factor authentication for voice purchases (EU compliance).
- Built backend systems for limited-time promotions handling peak retail traffic.
- Developed customer-data auditing/deletion services for privacy compliance.
- Led security reviews and capacity planning for multiple distributed services.

Tech stack: Java, Ruby, TypeScript, React, AWS (DynamoDB, Lambda, SQS, SNS, S3)

JPMorgan Chase & Co.

Brooklyn, NY

Software Engineering Intern

Jun 2019 – Aug 2019

- Built full-stack monitoring platform (React/Redux/Node.js) automating internal workflows; presented to leadership.
- Designed MariaDB schema and RESTful APIs powering real-time uptime dashboards and long-term analytics.

Tech stack: JavaScript, React, Redux, Node.js, MariaDB, REST APIs

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Master of Computer Science

May 2026 – May 2029 (Expected)

Stevens Institute of Technology

Hoboken, NJ

Bachelor of Engineering in Computer Engineering, Minor in Computer Science; GPA: 3.72

Aug 2016 – May 2020

TECHNICAL SKILLS

Languages: Python, TypeScript, C/C++, Java, JavaScript, Ruby, SQL, Bash

Embedded & ML: Zephyr RTOS, ARM Cortex-M, DSP, On-Device ML, Real-Time Systems, Bluetooth/BLE, ARM FVP

AI & LLM: RAG (embeddings, pgvector, hybrid search), LLM agents & tool calling, LLM evals, Claude Code, Codex

Backend & Data: Distributed Systems, PostgreSQL, DynamoDB, MariaDB, REST APIs, React, Node.js

Infrastructure & Tools: AWS (EC2, Lambda, S3, SQS, SNS, VPC), CloudFormation/CDK, Docker, Kubernetes, Git